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A Documentalist Critique of Information Science

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A Documentalist Critique of Information Science

Cover Page Footnote

I gratefully acknowledge the assistance of Ronald E. Day and Wayne de Fremery.

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This paper addresses the claim that new developments in document theory, sometimes referred to as Neodocumentalism, constitute a radical challenge to orthodox information science. I believe that the claim is valid with respect to information science *theory* (including library and information science *theory*), but I make no such claim with respect to practical information services. I do believe that the claim is correct with respect to theory, that it is a strong claim, and this is a significant change from earlier forms of documentation theory.

Comments on Information Science Terminology

Some clarification of what is meant by Information Science is needed. First, the term is widely used by two fields that overlap little in practice: There is an area commonly known as "library and information science" (LIS) which is centrally concerned with influencing what humans know and there is also the very important field of information science which, as is famously the case with Claude Shannon's communication theory, is not concerned with meaning. For the purposes of this discussion, I am only concerned with the former: Information science *with meaning*.

However, this qualification is not sufficient because of the ambiguity of the term "science." In stricter, British usage science *is* essentially theory. It comprises propositions to be refuted or qualified, so the phrase "science theory" is tautologous. In looser, US usage, "science" is freely used well beyond theory to include technical practices, as is terms like "domestic science" and "library science." Our purpose here is not to argue for any particular usage, but to emphasize that there is a pervasive ambiguity and to stipulate that this paper is concerned with theory as distinct from practice in the area commonly referred to as information science or library and information science (LIS). These distinctions suggest a basis for a language explanation for why theory has had a hard time developing within LIS.

Language Efficiency

The use of language tends towards efficiency. It should be effective with least effort. In practical matters, such as management, engineering, and everyday affairs, it is sufficient and efficient to use approximations and simplifications. There is limited need for precision to achieve effect. Also, the use of figurative language (analogies, metaphors, and similes) is very efficient and effective as an aid to rapid comprehension. Paul Otlet made constant and enthusiastic use of analogies and symbolism (Buckland, 2012). He was a rather extreme case but not untypical.

But in theory development, the deliberate distortions of imagery and figurative language become hindrances to efficiency. For theory development, rigor, and the simple, literal use of language is needed. This important difference has been noted by a few, but widely ignored. Building *theory* on the language of *practice* is a recipe for failure. It is ordinarily fine to say, for example, that a book is "relevant". But any such statement is highly problematic from a theoretical point of view. Relevance, by agreement, is a notion of central importance for information science but it has famously resisted satisfactory definition.

This is not simply a matter of Otlet's exuberance. The entire orthodoxy of library and information science is dominated by practical concerns. It is full of figurative language – and underdeveloped theory. A specific example is the idea that documents *represent* reality, like shadows in Plato's cave. In this view, documents are not reality but represent reality. And so you can change the format of documents and they can still represent reality. This leads to ideas of meaning as "content" that can be moved from one container to another and still represent reality, as in Otlet's famous drawings in his *Traité de documentation* (1934). However, this view is not unique to Otlet. It pervades all of LIS. The ASIST website quotes three widely respected, canonical definitions of information science. All three depend for intelligibility on the content/container concept which is a useful figurative metaphor that is untenable as credible theory. Pragmatically, the meaning is clear for everyday purposes. But as theory it is nonsense and a questionable choice for an academic professional association.

From Information to Document

A benefit of documentation studies has been to encourage attention to documents themselves rather than the hopelessly vague and over-used term *information*. What is a document? What could be a document? What do documents do? Niels W. Lund, Bernd Frohmann, Ron Day, and others have examined these questions. In effect, they have treated documents *as reality*, rather than *as representations of reality*. When this is done, there is suddenly a Copernican revolution: from Plato and the shadows in his cave to Aristotle and a more rigorous, more realist theory. By treating documents themselves as the reality, progress can be made if simplicity and rigor are not compromised by metaphor and figurative language.

How does all of this relate to Briet's antelope? It reminds us that the unfortunate antelope was part of reality. It was a unique living subject: a "particular". It had a mother, a birthdate, and unique childhood experiences. No other antelope could ever be exactly the same. However, every unique particular has features and, if any such feature is of interest to an observer, then the antelope can also be regarded as

a specimen exhibiting that feature. In philosophical terms, the consequence is simple. Ontologically, the material antelope *is* reality and, epistemically, a phenomenological view appears sufficient. And with that long introduction, I will resume where I ended at last year's Document Academy Annual Meeting, with two linear models, one of reading, and one of writing, developed by Wayne de Fremery and myself (Buckland & de Fremery, 2024)

Documents and their Features

Documents – etymologically objects that we may learn from – are complex objects with any number of physical features and cultural associations, far more than we could ever fully know or even list. We may think of documents and talk about documents as single, singular objects, treating them as nominal objects, things we talk about. But we do not engage with documents as whole objects. Instead, we engage with such features as we can sense and that interest us. It is a simplification to say that we observe a document. More precisely sense one or more *features* of a document and we try to understand – to make sense of – what we observe. There is no assurance that we perceive accurately, that we interpret correctly, draw plausible conclusions, or change our awareness of the world in a rational way. So our reading or reception of a document can be represented thus:

Document – Feature(s) – Reception – Awareness

It could be added that documents exist within contexts which influence both the choice of documents and their significance.

Context -- Document – Feature(s) – Reception – Awareness

That simple model represents the use of documents, not their origin. An analogous model of the creation of documents as originating in someone's mind, leading to some form of expression whose features constitute a document whose new existence becomes part of and thereby alters the broader context can be shown thus:

Awareness – Expression – Features – Document – Context

These two models can be dubbed, loosely, reception and expression respectively – or, more simply, reading and writing.

Their form is linear but linear here means "in a line" and does not necessarily mean direction. Rather, these linear representations are meant to show connectivity between the elements, adjacency, and where the elements are entangled with each

other. Consider, for example, the *reception* of a *feature*. Does the physical radiance of the feature denote direction toward the subject? Or, does the regard and interpretation by the subject imply direction from the subject to the feature? Is it both? The simplest solution is to describe it as an entanglement.

We can make two further modifications. First, awareness can be restated as the internal context of the mind in contrast to the external context of documents. Second, if both reception and expression are both abstracted to forms of cognitive construction, while remaining agnostic concerning the nature of cognitive activity. With these two modifications we find that the two linear models reduce to one unified model:

External context -- Document – Features – Construction – Internal context

And

Internal context – Construction – Features – Document – External context

They are the same model!

Comments on "Construction"

No particular theory of cognition is intended, least of all computer science related "cognitive science" – nor Dervin-esque "sensemaking" with its "gaps." But texts are interpreted and meaning is construed somehow. There seems no need here to make any particular assumptions about how it is done. Similarly, creating expressions is some form of construction but, equally, I do not see the need to assume any particular process. All that said, some term is needed, so unless better emerges, I am inclined to use "cognitive construction" – with disclaimers.

"Public Knowledge"

"Public" is a convenient abstraction for a group of people. It is a simple way of referring to many people within some population. So public knowledge is shorthand for saying that many individuals know, or, as I would prefer, believe.

Documentation as Context Engineering

Documentation can be understood as generating changed context is two quite different ways. As we try to cope we need to somehow reconcile (harmonize) our perceived external context and our internal context. We can do this by adjusting our

beliefs to what we perceive and/or changing the perceived external context, by, for example, selective attention, a changed perspective, generating new evidence (by experiment or faking) – or wishful thinking. Also, the perceived external context is continuously harmonized with the extant context (the Universe) through discovery and experiments.

1. Rearrangement

Context can be transformed without generating any new documents simply by rearrangement. Information retrieval discourse frequently invokes the proverbial needle in haystack. The reality is that finding a particular document or any document with some unusual feature is easy if it is in a very small collection but impractical in any very large collection, especially one with millions of documents unless the collection has been arranged to facilitate finding it. The idea is that finding a needle in a haystack would be comparably difficult. In both cases finding the sought object is certainly possible, and, in theory, one might be lucky and find it quickly. But that is very improbable and the expected amount of labor required would be excessive. But document retrieval systems are designed to transform the situation by moving the desired items from a very large collection to a very small collection -- the retrieved set – where it will be one among a dozen instead of one among millions. It is comparable to moving the needle from the haystack and putting it in a pincushion where it is very easily and quickly found. Of course, collections of documents are rarely without some kind of organization. The problem is that reasons for seeking document vary depending on the feature or aspect that is of interest and any given arrangement is likely to be unsuitable for the current search need. Arrangements by size, order received, size, or date of creation would be unhelpful for finding a particular book if only the title is known or any book about a specific topic.

2. Generating Additional Context

Bibliographical description supplements whatever is described. That description can be searched and, thereby, is used as context. Bibliographical description is a special case of the more general category of metadata. Metadata, therefore, can be understood as additional context.

Informal terms for these two different but complementary processes are parking and marking.

Metadata – data around or beyond something – is essentially description. (Metadata relating to documents is bibliographical description.) Metadata is enabling infrastructure. Infrastructure is context.

Disclaimer on Language Use

We started with a critique of language usage in information science as a hindrance to theory development. But these are not new claims. It is now twenty-five years since Ron Day completed his pioneering critique *The modern invention of information: Discourse, history, and power* (2001). It is detailed historical analysis of how the dominant ideological theme of modern society was crafted without any credible scientific basis to advance a wide range of vested interests. The "information society" is revealed as a literary concoction, not a scientific one. The first few chapters provide a very readable introduction. Day has continued to explore these issues in many later publications.

Shortly thereafter, after many years of intense research, Bernd Frohmann completed his important book, *Deflating Information: From science studies to documentation* (2004). Frohmann explained that the concepts and rhetoric dominating library and information science were fundamentally flawed by treating "information" as an imaginary, non-material "epistemic content" that resists critical analysis. Frohmann's book remains valid. It is very carefully written, but not an easy read.

These two books are salient, other writings with similar tendencies can be found primarily in the renewal on interest in documentation theory since the mid-1990s.

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